

Classification of Motor Imagery EEG Signals Based on Wavelet Transform and Sample Entropy

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Abstract—Modeling and learning of brain activity patterns represent a huge challenge to the brain-computer interface(BCI) based on electroencephalography(EEG). Many existing methods based on EEG signal frequency domain feature extraction can not achieve high classification accuracy rate requirements[1]. This paper presents a method of feature extraction of EEG based on wavelet transform and sample entropy. The dynamic changes of the sample entropy of the left and right hand motion imaginary EEG signal and its neurophysiological significance were analyzed. Finally, Fisher's linear discriminant was used to classify left-right hand motion imaginary EEG. Experiments shows that the average correct rate of classification was 88.9%.

Keywords—brain computer interface(BCI); motor imagery(MI); Wavelet transform; sample entropy(SampEn)

I. INTRODUCTION

BCI is a system that allows the human brain to communicate directly with the external interface[2]. The traditional exchange and control between the human brain and the external environment relies on peripheral nerves and muscle tissue, and BCI provides a direct communication path between the human brain and the external device. Real-time control of the equipment is realized by real-time measuring the brain activity related to the user's intention and converting the activity into the corresponding control signal[3~4](Fig. 1 for the basic BCI system block diagram). BCI ultimate goal is to form a more natural smooth man-machine communication. This can increase the personnel's special control skills for special equipment, such as divers, astronauts, etc., in certain special environments, as well as reducing personnel workload and improving work efficiency and control accuracy.

The most important part of realizing BCI system based on left-right hand motor imagery is the feature extraction of EEG signal. The current feature extraction method is mainly based on left-right hand motor imagery EEG different frequency bands of energy changes. When a person in the unilateral limb movement to imagine the contralateral side of the brain μ rhythm and β rhythm will appear event related desynchronization(ERD) phenomenon, at the same time, event related synchronization(ERS) occurs on the same side[5~6]. AR model and power spectrum estimation only analyze the frequency

domain characteristics of EEG signal, while neglecting the time domain characteristics. These feature extraction methods can not guarantee the correct classification Therefore, looking for more effective left-right hand motor imagine EEG features for the improvement of BCI performance is very meaningful.

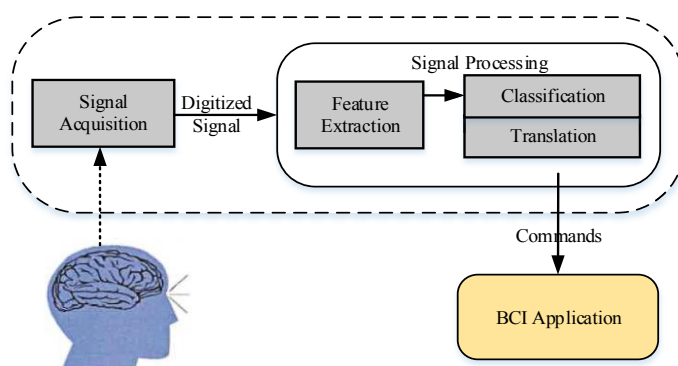


Fig. 1. Basic BCI system block diagram

In this paper, wavelet and sample entropy were combined to extract EEG signal feature. In the left-right hand motor imagery EEG classification problem, sample entropy as classification feature has the following advantages:

- SampEn can measure the probability of generating new patterns in time series. The larger SampEn value, the greater the probability that the new pattern will be generated, resulting in the more complex sequence[7], and this can be used to measure changes in the complexity of the EEG signal when the brain sensory motor cortex is activated during motion imaging.
- SampEn does not contain the comparison of its own data segment, which is the exact value of the negative average natural logarithm of the conditional probability. The calculation does not depend on data length and the computation is small. This is conducive to improve the system real-time with less BCI training time.
- SampEn is not sensitive to the data loss, and it is with a good anti-jamming capability. This is very useful for analyzing EEG signals with very low signal-to-noise ratios, which can enhance the usefulness of BCI.

In the experiment, firstly, the EEG original signal is denoised by wavelet transform, then the nonlinear dynamical parameter "sample entropy" is used to classify EEG features. The dynamic changes of sample entropy of left-right hand motor imagine EEG signal and its neurophysiological significance were analyzed. Finally, Fisher's linear discriminant was used to classify left-right-hand motion imaginary EEG. The experiment achieves a good classification effect, which proves that the scheme has great feasibility and practical value.

II. FUNDAMENTALS OF ALGORITHMS

A. The principle of wavelet decomposition

Wavelet transform is a time-frequency analysis method, can be a good characterization of the signal in time domain and frequency domain characteristics[8]. Wavelet transform can be used to decompose the EEG signals containing noise into multiscale decomposition, and then the subband signal containing noise is processed, so as to achieve the purpose of signal denoising.

In practical applications, it is common to use the dynamic dyadic wavelet for the discretization of the scale and the discretization of the displacement (the displacement should be chosen so as not to lose the information). The definition of discrete wavelet transform:

$$WT_f(j, k) = \int f(t) \psi_{j,k}^*(t) dt \quad (1)$$

$$\psi_{j,k}(t) = 2^{-\frac{j}{2}} \psi(2^{-j}t - k) \quad (2)$$

Discrete wavelet inverse transform is

$$f(t) = \sum_{j,k} \langle f, \psi_{j,k} \rangle \cdot \tilde{\psi}_{j,k}(t) = \frac{1}{A} \sum_{j,k} WT_f(j, k) \cdot \psi_{j,k}(t) \quad (3)$$

When j, k is different, the wavelet transform decomposes the signal $x(t)$ into different subbands, j modulates the frequency domain localization of the signal, k modulates the time domain localization of the signal. When the value of j is large, the low-frequency characteristic of the signal is analyzed; when j is small, the high-frequency characteristic of the signal is analyzed. In practical applications, wavelet decomposition levels are determined from the range of useful signals and noise bands. Using the wavelet transform formula (1), the original EEG signal is decomposed into different frequency range of the signal, And then the band containing noise is processed, Finally, according to formula (3) we reconstructed the denoised EEG.

B. Fundamental theory of sample entropy

SampEn is a new measure of time series complexity proposed based on ApEn. It solves the problem of inconsistency of statistics in ApEn and is therefore more accurate than ApEn[9].

SampEn's algorithm definition: For a given time series of $\{u(i)\}$ containing N points, the calculation of SampEn is as follows (m is a preselected pattern dimension, r is a preselected similarity margin):

1) Orderly composing the sequence $\{u(i)\}$ into m dimension vector:

$$X_m(i) = [u(i), u(i+1), \dots, u(i+m-1)], i = 1, 2, \dots, N-m \quad (4)$$

2) For each i value, the distance between the vector $X_m(i)$ and the rest vector $X_m(j)$ is calculated accordingly:

$$d[X_m(i), X_m(j)] = \max_{k=0 \sim m-1} |u(i+k) - u(j+k)| \quad (5)$$

($i, j = 1 \sim N-m, i \neq j$)

3) Given a similar margin r value ($r > 0$), conducting the number of statistics $d[X_m(i), X_m(j)] < r, (i, j = 1 \sim N-m, i \neq j)$ for each i value, and denoting the ratio of this number to the total number of distance $N-m-1$ as $B_i^m(r)$:

$$B_i^m(r) = \{d[X_m(i), X_m(j)] < r, (i \neq j)\} / (N-m-1) \quad (6)$$

4) Finding the average of $B_i^m(r)$ for all i value to denote as $B^m(r)$:

$$B^m(r) = \frac{1}{N-m} \sum_{i=1}^{N-m} B_i^m(r) \quad (7)$$

5) Adding dimension as $m+1$ to construct $m+1$ dimension vector:

$$X_{m+1}(i) = [u(i), u(i+1), \dots, u(i+m)], i = 1 \sim N-m \quad (8)$$

6) For each i value, calculating the distance between the vector $X_{m+1}(i)$ and the rest vector $X_{m+1}(j)$:

$$d[X_{m+1}(i), X_{m+1}(j)] = \max_{k=0 \sim m} |u(i+k) - u(j+k)| \quad (9)$$

($i, j = 1 \sim N-m, i \neq j$)

7) For the number of each i value, conducting the statistics for the number of $d[X_{m+1}(i), X_{m+1}(j)] < r, (i, j = 1 \sim N-m, i \neq j)$, and the ratio of this number to the distance of total number $N-m-1$, denoting as $B_i^{m+1}(r)$:

$$B_i^{m+1}(r) = \{d[X_{m+1}(i), X_{m+1}(j)] < r, (i \neq j)\} / (N-m-1) \quad (10)$$

8) Finding the average of $B_i^{m+1}(r)$ for all i value, denoting as $B^{m+1}(r)$:

$$B^{m+1}(r) = \frac{1}{N-m} \sum_{i=1}^{N-m} B_i^{m+1}(r) \quad (11)$$

Theoretically, the sample entropy of the sequence is:

$$\text{SampEn}(m, r) = \lim_{N \rightarrow \infty} \left\{ -\ln \left[B^{m+1}(r) / B^m(r) \right] \right\} \quad (12)$$

When N is a finite value, getting the estimated value of SampEn when the sequence length is N , denoting as:

$$\text{SampEn}(m, r, N) = -\ln \left[B^{m+1}(r) / B^m(r) \right] \quad (13)$$

Of course, the value of SampEn is related to the value of m , r , according to experience, $m=2$ is usually taken, $r=0.1 \sim 0.25SD(u)$, wherein, $SD(u)$ is expressed as the standard deviation of the original time series.

III. EXPERIMENT AND RESULT ANALYSIS

A. Experimental data

Experimental data Using the 2008 Brain-Computer Interface Competition provided Data sets 1 calibration data for subject b. The experiment uses 59 electrodes, the sampling frequency is 100Hz, the period is 8s. The subjects sat relaxed in the comfortable seat in front of the computer. According to the prompts on the screen to carry out sports imagination. As shown in Fig. 2, each experimental cycle is divided into the following steps:

- 1 ~ 2s showing a black screen state, the subjects remain quiet and relaxed;
- When the time to 2s, the computer issued a brief beep, to remind the subjects of attention;
- 2 ~ 4s appears on the screen cross '+', so that the subjects focus on the center of the screen;
- When to 4s, the screen appears to the left or right arrow, Subjects under the direction of the arrow to the left hand or right hand movement imagination, sustained 4s.

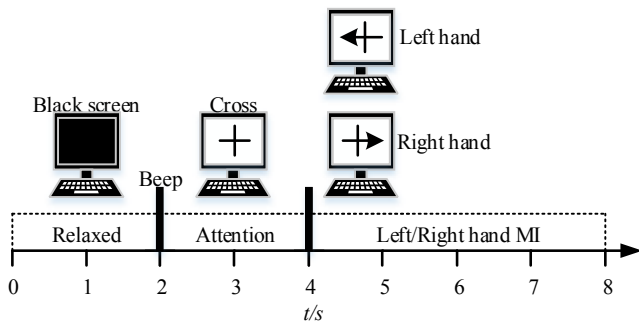


Fig. 2. Experimental process timing diagram

The experimental data included left and right hand motor imagery of the subjects for 100 times. The first 50 data were

used for training, and the last 50 were for testing. Fig. 3(a) for the brain functional area distribution, Fig. 3(b) is part of the electrode channel distribution.

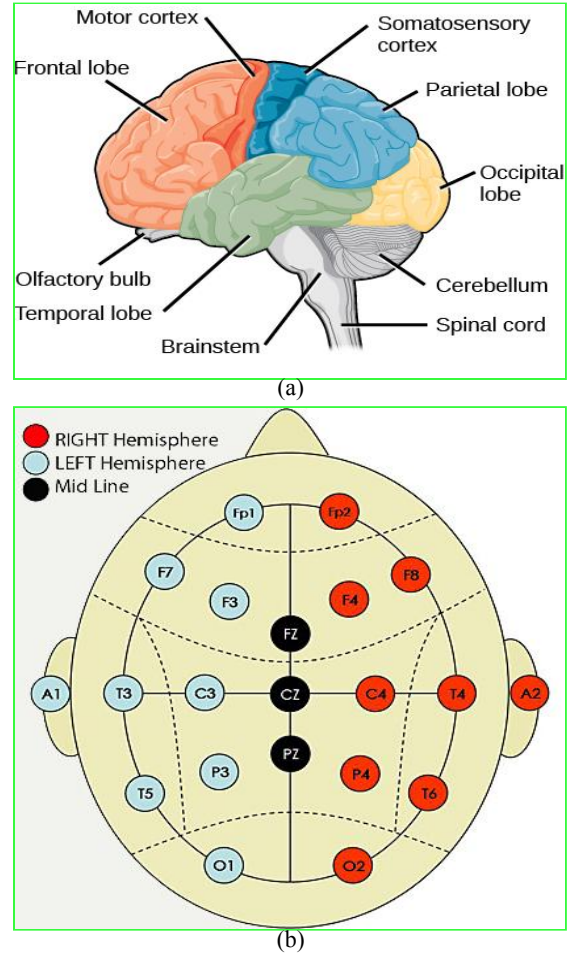


Fig. 3. Brain function area and part of the electrode distribution

B. Motor imagery EEG signal preprocessing

Since the EEG signal is very weak (μV level), and internal mixed with a variety of noise (such as EOG, EMG, ECG, Power Frequency Noise, etc.) [10~11]. So the signal to noise ratio of EEG is very low. In order to reduce noise, we use the nonlinear wavelet transform threshold method to process the original EEG signal (Fig. 4 is the original EEG signal of two experimental periods). Through experimental simulation, select db4 wavelet decomposition of 5 layers. Fig. 5 shows the decomposition of the low-frequency details. Found that useful EEG signals are concentrated in 1 to 3 layers. The high frequency coefficient above the fifth layer is directly set to 0, and then use this soft threshold method to deal with the signal of Fig. 4. The denoised signal is shown in Fig. 6.

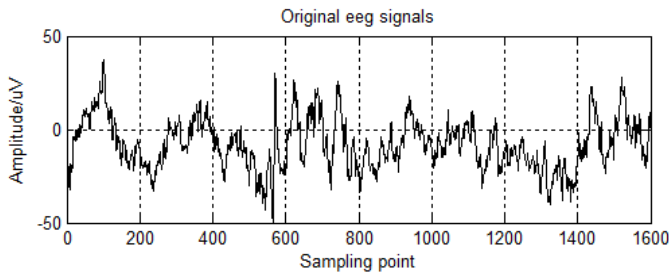


Fig. 4. Contains the noise of the EEG signal

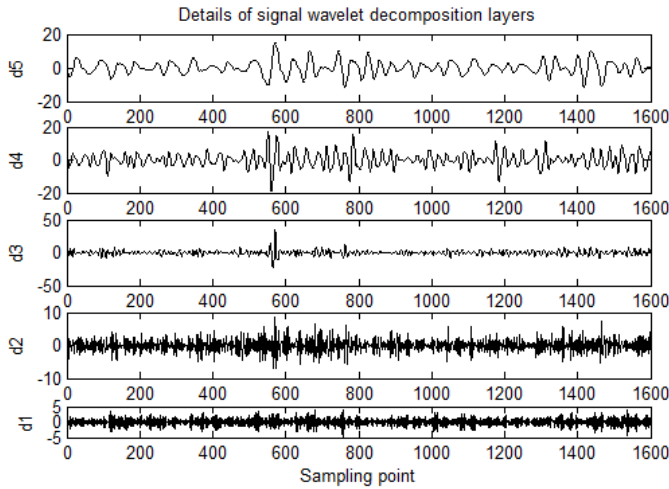


Fig. 5. Low-frequency detail of EEG decomposition

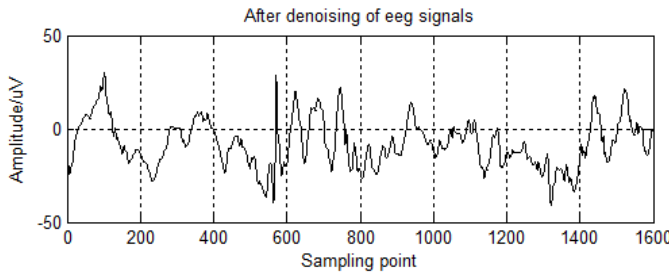


Fig. 6. Soft threshold method for denoising EEG signals

As can be seen from Fig. 4, the original EEG signal is not smooth, with significant high-frequency noise, which will directly affect the waveform characteristics of the extraction. From the above simulation results can be seen in Fig. 6: After filtering the signal to filter out noise interference. And retains the EEG information, which provides powerful guarantee for the feature extraction and classification recognition.

C. Complexity analysis of EEG during left-right hand motor imagery based on sample entropy

When people conduct motor imagery, there will be 8-12Hz μ rhythm and 18-26Hz β rhythm EEG generated in the brain sensory motor cortex area[12~13]. When people conduct the unilateral limb motor imagery, the contralateral μ rhythm and β rhythm of the brain will appear amplitude attenuation (ERD); and the ipsilateral μ rhythm and β rhythm of the brain will appear amplitude enhancement(ERS). Based on ERD and ERS phenomena, We can quantify the complexity of the ERD time-series by calculating the entropy phase of the EEG sample during left-right-hand motor imagery.

Main steps are as follows: First, by starting from the first second of each experimental period, setting the width of sliding time window as 1s (i.e. 100 points), calculating the SampEn of the EEG signal of C3 and C4 channels 1s before the sampling point. The window moves backward a sampling point for each time, until the SampEn of the last data is calculated, thus to obtain the time series of SampEn for the EEG signal of C3 and C4 channels of an experimental period. Then the SampEn sequences of the same experimental data of motor imagery are superimposed and averaged, thus to obtain the average SampEn ERD curve of a single subject. Finally, the SampEn ERD curves of different subjects are superimposed and averaged, thus to obtain the final statistical SampEn ERD curve, which represent the variation of EEG complexity over time. Left and right hand motor imagery SampEn ERD curve is as shown in Fig. 7 and Fig. 8.

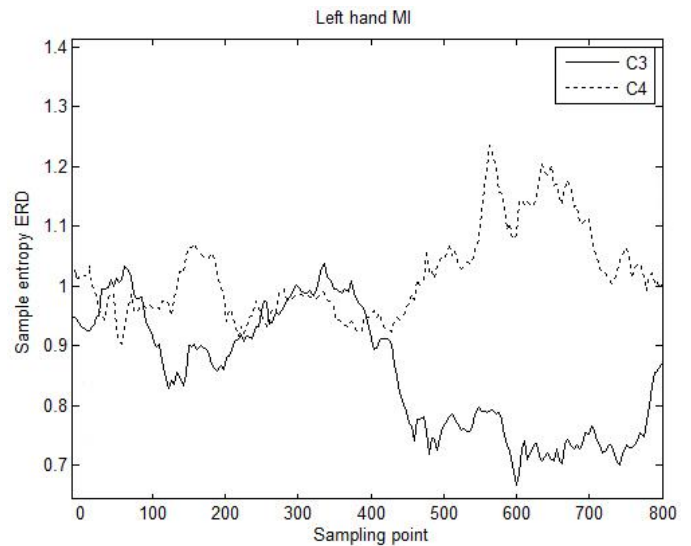


Fig. 7. Left hand motor imagery SampEn ERD curve

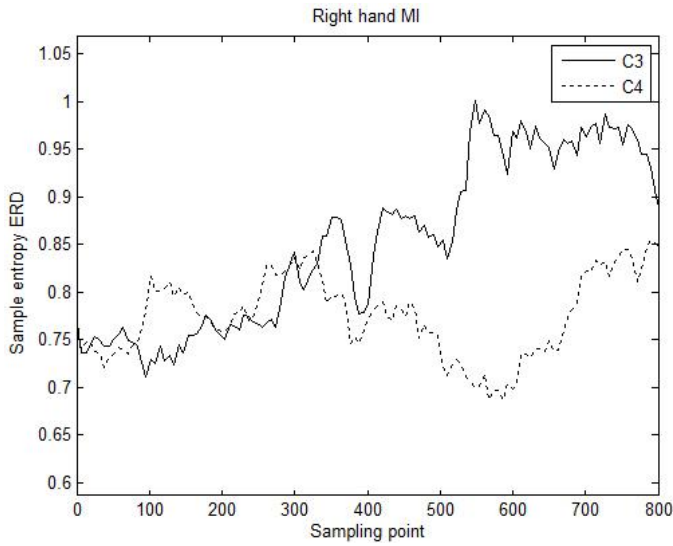


Fig. 8. Right hand motor imagery SampEn ERD curve

Seen from the Fig. 7 and Fig. 8 that when the subjects conduct left hand motor imagery, the EEG signal SampEn of the same side C3 channel of the brain reduces significantly, in other words, the complexity reduces significantly, and EEG signal SampEn of the other side C4 channel of the brain increases slightly, that is to increase complexity. When the right hand movement imagination of the experimental results and the left hand is just the opposite.

According to the neurophysiology, when people conduct the left-right hand motor imagery, the sensory motor cortex area of the contralateral brain activates and the activity of a large number of neurons enhances accordingly. Meanwhile information processing leads to the enhancement of electroencephalogram signal independence and the decrease of the synchronization, thereby resulting in amplitude attenuation (ERD) of μ rhythm and β rhythm, however, there is the phenomenon of the increased electroencephalogram complexity. The ipsilateral brain sensory motor cortex area is relatively static, and the inhibition of neuronal activity causes the increase of electroencephalogram signal synchronization, resulting in μ rhythm and β rhythm amplitude increase (ERS), electroencephalogram complexity decreases. Based on the above analysis, it is obvious that SampEn complexity of the left-right hand motor imagery EEG has the same physiological basis as its own energy, which proves that it is feasible to use SampEn as the EEG complexity feature to classify the left-right hand motor imagery tasks.

D. Feature extraction and classification

In this study, SampEn time series of EEG signals of $SampEnC3(t)$ and $SampEnC4(t)$ of C3 and C4 channels in a single experiment obtained through calculation combined into two-dimensional time-varying feature vectors

$Ft = (SampEnC3(t), SampEnC4(t))$ for real-time continuous classification of motor imagery tasks. The SampEn of 1s before the EEG signal calculated as the SampEn of the EEG signal, and the similar margin r calculated by taking $0.15SD(u)$.

The Fisher linear discriminant used to classify left-right hand motor imagery tasks for dynamically continuous classification to obtain good results. Firstly, the time-varying feature vectors of the training samples and the test samples are calculated. The weight coefficient $W(t)$ and threshold ω_0 of the Fisher discriminant at each sampling point obtained from the feature vectors of training sample. Then is to calculate Fisher discriminant distance for each sampling point feature vector of the test sample, determining their classification, thus to achieve continuously dynamic classification.

The linear discriminant function is:

$$g(x) = W^T x + \omega_0 \quad (14)$$

Wherein, x is the tested feature vector, when $g(x) > 0$, the decision x belongs to the first class (left hand imagination), when $g(x) < 0$, the decision x belongs to the second class (right-hand imagination); when $g(x) = 0$, the decision x does not belong to any class.

Using the Fisher criterion, calculating the formula W :

$$W = (S_L + S_R)^{-1} (u_L - u_R) \quad (15)$$

Wherein S_L , S_R are respectively the intra-cluster divergence matrices of left-right hand imagery training eigenvectors, u_L , u_R are respectively the mean values of the eigenvectors of left-right hand motor imagery training eigenvectors.

ω_0 value estimated by the following formula depends on experience:

$$\omega_0 = -\frac{W^T (u_L + u_R)}{2} \quad (16)$$

Variance EEG SampEn serves as eigenvector when using two-dimension, and using Fisher's linear discriminant to classify results. The result of each classification obtained by superimposing the results of different subjects, and finally is for the taking the average classification accuracy rate of up to 88.9%. The experimental results show that SampEn can characterize the EEG complexity of left-right hand motor imagery, and there is good effect on classification.

IV. CONCLUSION

This paper takes SampEn as a feature vector for making the research on the dynamic classification of the left-right hand motor imagery task, achieving good results, which offers a new

way of thinking for the BCI feature extraction. Experiments show that SampEn can represent the complexity change of EEG, there is a clear physiological significance, but also there are significant differences in the left-right hand motor imagery, which is in favor of BCI classification accuracy improvement, and is also a very useful value of the characteristic parameters.

In this study, the sample entropy method is applied to BCI real-time system based on left-right hand motion imagination. To achieve the brain directly control the robot forward and stop. The experiment has achieved good results. The next step is to combine the EEG frequency domain features and the complexity characteristics represented by the sample entropy. Prediction, experiments can get better classification results.

REFERENCES

- [1] Daly,L, Muller-Putz,G, et al. Fully Online and Automated Artifact Removal for Brain-Computer Interfacing[C]. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2015, 23(5): 725-736.
- [2] Ye Liu ;Dept. of Comput. Sci. ; Eng., Shanghai Jiao Tong Univ., Shanghai, China;Hao Zhang;Min Chen;Liqing Zhang . A Boosting-Based Spatial-Spectral Model for Stroke Patients' EEG Analysis in Rehabilitation Training. Neural Systems and Rehabilitation Engineering, IEEE Transactions on. 2016. 24(1): 169-179.
- [3] S. Saeedi; R. Chavarriaga; R. Leeb; J. d. R. Millan. Adaptive Assistance for Brain-Computer Interfaces by Online Prediction of Command Reliability. Computational Intelligence Magazine, IEEE. 2016. 11(1): 32-39.
- [4] V. Jayaram; M. Alamgir; Y. Altun; B. Scholkopf; M. Grosse-Wentrup. Transfer Learning in Brain-Computer Interfaces. Computational Intelligence Magazine, IEEE. 2016. 11(1): 20-31.
- [5] Nicolas-Alonso, L.F. ;Biomed. Eng. Group, Univ. de Valladolid, Valladolid, Spain ; Corralejo, R.; Gomez-Pilar, J.; Alvarez, D. more authors . Adaptive Stacked Generalization for Multiclass Motor Imagery-Based Brain Computer Interfaces. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2015. 23(4): 702-712.
- [6] Lianghua He; Die Hu; Meng Wan; Ying Wen; Karen M. von Deneen; MengChu Zhou. Common Bayesian Network for Classification of EEG-Based Multiclass Motor Imagery BCI. Systems, Man, and Cybernetics: Systems, IEEE Transactions on. 2016. 46(6): 843-854.
- [7] Mourad Kedadouche; Marc Thomas; Antoine Tahan; and Raynald Guilbault. Nonlinear Parameters for Monitoring Gear: Comparison Between Lemple-Ziv, Approximate Entropy, and Sample Entropy Complexity. Shock and Vibration. 2015. 2015(000): 1-13.
- [8] Dongying Han; Pei Li; Shujun An; Peiming Shi. Multi-frequency Weak Signal Detection Based on Wavelet Transform and Parameter Compensation Band-pass Multi-stable Stochastic Resonance. Mechanical System and Signal Processing. 2016.70-71(000): 995-1010.
- [9] Lei Zhang; Long Zhang; Junfeng Hu; and Guoliang Xiong. Bearing Fault Diagnosis Using a Novel Classifier Ensemble Based on Lifting Wavelet Packet Transforms and Sample Entropy. Shock and Vibration. 2016. 2016(000): 1-14.
- [10] Bermudez i Badia, S.[1]; Using a Hybrid Brain Computer Interface and Virtual Reality System to Monitor and Promote Cortical Reorganization through Motor Activity and Motor Imagery Training. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2013. 21(2): 174-181.
- [11] Fabian Chersi[1];Francesco Donnarumma;Giovanni Pezzulo; Mental imagery in the navigation domain: a computational model of sensory-motor simulation mechanisms. Adaptive Behavior . 2013. 21(4): 251-262.
- [12] Pengfei Wei[1];Wei He;Yi Zhou;Liping Wang. Performance of Motor Imagery Brain-Computer Interface Based on Anodal Transcranial Direct Current Stimulation Modulation. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2013. 21(3): 404-415.
- [13] Bin He ;Dept. of Biomed. Eng., Univ. of Minnesota, Minneapolis, MN, USA ; Baxter, B.; Edelman, B.J.; Cline, C.C. more authors. Noninvasive Brain-Computer Interfaces Based on Sensorimotor Rhythms. Proceedings of the IEEE. 2015. 103(6): 907-925.